

1. Stigmata or spiracles: The tracheal system communicates with the exterior by ten pairs of openings called **stigmata** or **spiracles**. The first two pairs of spiracles are present in the thoracic segments, one pair in the mesothorax and one pair in the metathorax. The remaining eight pairs are present in the first eight abdominal segments. Spiracles are located in the pleura of their respective segments. The respiratory system in insects is classified on the basis of number and nature of spiracles. The respiratory system of cockroach is **polypneustic** (as there are more than 3 pairs of spiracles) and the spiracles are **holopneustic** (as all of them are functional). All spiracles are valvular and each of them is surrounded by a chitinous ring called **peritreme**. All spiracles bear small hair like structures called **trichomes** to filter the dust particles. Each spiracle opens into a small chamber called **atrium**.

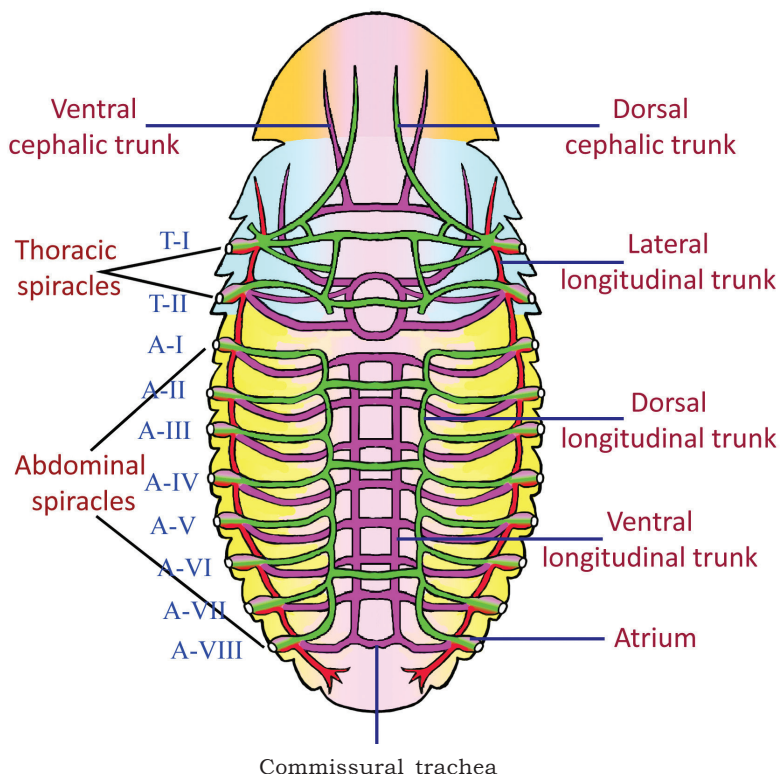


Fig. 7.7 *Periplaneta* : Tracheal system

2. Tracheae: From the atrium of each **thoracic spiracle** several horizontal tracheae run inside. They join with each other in the thorax to form many tracheal trunks like dorsal cephalic, ventral cephalic trunks and their branches. These branches enter all organs of the head. The thoracic region contains lateral longitudinal trunks also. The **abdominal spiracles** lead into atria. From the atrium of each abdominal spiracle three tracheal tubes arise. All these tracheal tubes of one side open into three separate longitudinal tracheal trunks. They are **lateral**, **dorsal** and **ventral longitudinal trunks**. Lateral longitudinal trunks are the longest tracheal trunks. The three pairs of longitudinal tracheal trunks of both the sides are interconnected by many **commissural tracheae**. From all the tracheal trunks several branches are given out, and they enter different organs. Each tracheal branch ends in a special cell called **tracheole cell**.

The wall of the tracheae is made of three layers. They are an outer **basement membrane**, a middle one cell thick **epithelium** and an inner layer of cuticle called **intima**. The intima is produced into **spiral** thickenings called **taenidia**. The taenidia keep the tracheae always open and prevent it from collapsing.

2. Tracheoles

The terminal cell of trachea is called **tracheoblast** or **tracheole cell**. It has several intracellular tubular extensions called **tracheoles**. Tracheoles are devoid of intima and taenidia. They are formed of a protein called **trachein**. Tracheolar fluid is present inside the tracheoles. The level of the tracheolar fluid varies with the metabolic activity of the insect. It is more when the insect is inactive and completely reabsorbed into the tissues, when the insect is more active. Tracheoles reach tissues/ cells and are intimately associated with mitochondria of the cells (to supply oxygen to them).

Which structures prevent the tracheae from collapsing?

4. Mechanism of respiration

Respiration includes two events, viz., inspiration and expiration. The muscles helpful are **dorsoventral muscles** and **ventral longitudinal muscles**. Dorsoventral muscles are the principal muscles of respiration.

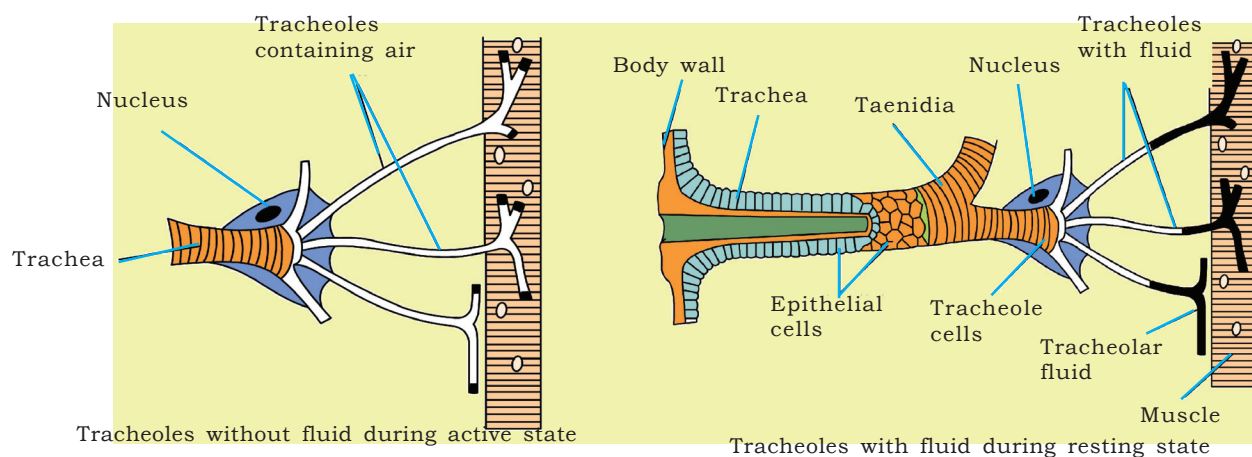


Fig. 7.8 *Periplaneta* : Tracheoles showing movement of tracheolar fluid during respiration

A. Inspiration

Taking in of air is called **inspiration**. It is effected by the relaxation of the dorsoventral muscles and ventral longitudinal muscles. Due to the relaxation of the dorsoventral muscles, tergal plates are elevated and the volume of the body cavity increases. Due to the relaxation of the ventral longitudinal muscles, the telescoped segments come to the **normal position**. So the volume of the body cavity increases in the longitudinal axis. As air is drawn in due to the relaxation of the muscles, the process is a 'passive' process. During inspiration the thoracic spiracles are kept open and the abdominal spiracles are kept closed.